

Unit: Unit 6: Systems of Linear Equations and Inequalities

Topic: Applications of Systems (Word Problems)

Name: _____

Date: _____

Foundation (Jalapeno)

Open Ended Questions (FRQ)

Q9. A company produces x units of Product A and y units of Product B. The profit on each unit of Product A is 10 and the profit on each unit of Product B is 15. The company has 24000 available for production and each unit of Product A costs 100 to produce and each unit of Product B costs 120 to produce. Write a system of equations to represent this situation and find the maximum profit.

Q10. A business owner has x dollars to spend on advertising and y dollars to spend on salaries. The total budget is 15000. If advertising costs 300 per ad, write an inequality to represent this situation and graph the solution on a coordinate plane.

Chef's Tips

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- Read each question carefully and choose the correct answer.
- Check your work and make sure your answers are reasonable.
- Use a calculator or graphing tool if necessary.